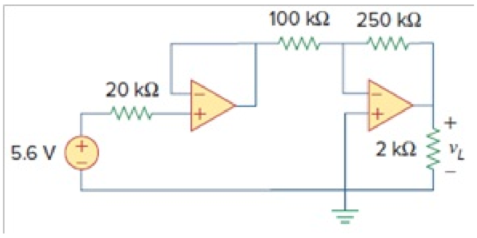
Chapter 5, Problem 72P

Problem

Find the load voltage *vL* in the circuit of Fig. 5.98.

Figure. 5.98:



Step-by-step solution

Step 1 of 2

Draw the circuit diagram as shown in Figure 1.

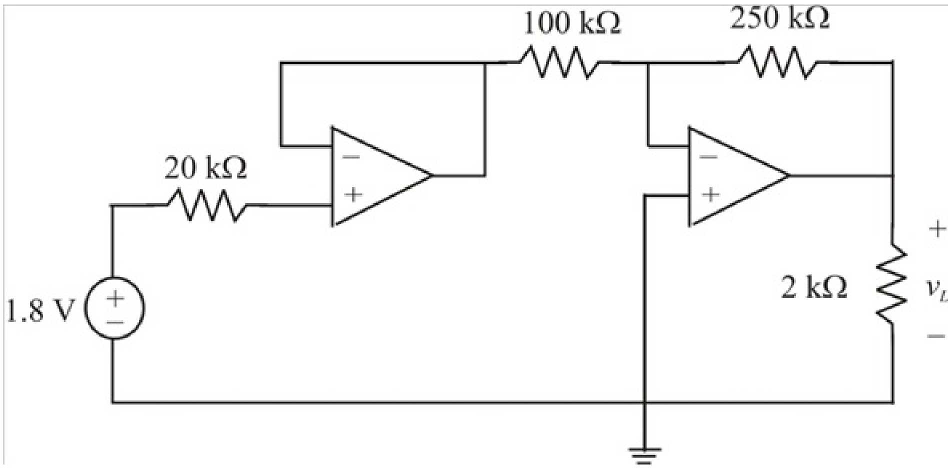


Figure 1

Step 2 of 2

Since no current flows into the input terminals of the ideal op amp, there is no current flow through the 20 k? resistor.

Hence the output voltage of the voltage follower (output of first op amp) is 1.8 V

The second amplifier is an inverting amplifier, the output voltage of an inverting amplifier is given by:

$$v_o = -\frac{R_f}{R_1} v_i$$

Where

$$R_f = 250\,\text{k}\Omega$$
$$R_1 = 100\,\text{k}\Omega$$
$$v_i = 1.8\,\text{V}$$

The output voltage v_L of the amplifier is calculated as:

$$\begin{aligned} v_L &= -\frac{250}{100}(1.8) \\ &= -(2.5)(1.8) \\ &= -4.5\,\text{V} \end{aligned}$$

Therefore the output voltage v_L is $\boxed{-4.5\,\text{V}}$.